

Parametric analysis of wind siting efficiency

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Abstract

The effect of wind climatic characteristics on the efficiency of the WECS is examined by means of the Weibull general model and a simple model of power curve. Two efficiency parameters—the plant utilization factor (F_u) and the site effectiveness (ε)—are considered as functions of three parameters: the Weibull shape parameter (k), a dimensionless mean wind speed (α), and the design ratio between rated and cut-in speed (ϕ). It is shown that only for intermediate values of α and ϕ the plant utilization factor can be considered independent of k , being otherwise either improved or penalized in no negligible measure by it, depending on the values of α and ϕ . The effect of increasing k on the site effectiveness is always beneficial. The effect of increasing α is beneficial for the plant utilization factor but largely penalizing for the effectiveness. However, no maximization is possible for neither of the efficiency parameters.

It is shown that the correct use of F_u and ε for comparison is significant only for a given WECS, while the WECS data have to be used when comparing different or modified wind machines. The data of 35 commercial WECS are examined to show that the common assumption of constant rated or maximum efficiency is not acceptable.

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1. Introduction

In the past, the wind electric conversion systems (WECS) were not adjustable machines, and the only choice one had was to choose among different models; as the wind energy resource can neither be controlled nor adjusted, the suitability of the machine to a given site was rather uncontrollable. The present growth of the wind

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Nomenclature

A	turbine disk area
C_p	power coefficient of the turbine
$C(V)$	cumulative function of the wind frequency distribution
D	turbine diameter
E	output energy of the WECS
E_a	available energy in the site, Eq. (1)
$E_{\max}$	maximum value of the output energy = $\eta_{\max} E_a$
E_r	nominal energy of the WECS = TP_r
$f(V)$	wind speed probability function
f_1, f_2	fractions of probable-time in the ranges 1 and 2
V	wind speed
V_c	cut-in wind speed
V_f	cut-off wind speed
V_r	rated wind speed
V_m	mean wind speed in the period T
$V_{m2,1}$	root-mean-square wind speed in the range 1, Eq. (10)
V_{m3}	root-mean-cube wind speed, Eq. (1)
v	dimensionless wind speed, related to the mean wind speed
P	output power of the WECS
P_a	available power in the disk area = $1/2\rho AV^3$
P_r	rated power of the WECS
t	time
T	period of time of analysis
α	intensity ratio = V_m/V_c
η	overall power efficiency of the WECS, Eq. (2)
η_{el}, η_m	electrical and mechanical efficiencies
$\eta_{\max}$	maximum WECS efficiency, Eq. (5)
η_r	rated WECS efficiency, Eq. (4)
ρ	air density
ϕ	design ratio = V_r/V_c
$\{i\}$	range i th of the power curve, Fig. 1

energy exploitation together with the industrial competition tend to make the WECS more similar to water turbines which can be adjusted to the resource. However, the question is: what is a satisfactory and exhaustive measure of the wind resource? Unlike the hydropower resource, which being continuative is fully defined by the water head, the wind resource depends not only on the wind speed but also on its distribution in the time—the wind frequency distribution. The unforeseeable irregularity of the wind does not allow to design fluid-dynamically optimized (i.e. working at maximum efficiency) turbines, as this requires constant or slowly and little varying fluid-dynamic conditions at inlet.

Overall optimization is possible over long periods by matching the wind frequency distribution with the power curve of the wind system, and the overall energy output can be maximized by means of a suitable choice of the power characteristics of the system. However, the term “optimization” has to be used in a general sense, as there is indefiniteness about the parameter to maximize and not all of them can be maximized. Generally speaking, we are interested in maximizing the energy output, but something must be held constant. All turbine optimization criteria consider a dimensionless form of the energy (efficiency) to be expressed analytically or numerically as a function of a dimensionless form of a design (usually kinematic) parameter of the turbine. The optimization consists in finding the optimum value of the design parameter, which is what maximizes the efficiency. In the case of the wind turbines, the long-term energy output can be related either to the rated energy of a given machine (that is the energy the machine should deliver if constantly operating at the rated power) or to the available wind energy of a given site; these parameters are the plant utilization factor (or capacity factor) and the energy efficiency, respectively. As it will be shown in this paper, these two quantities have different behavior and can never be maximized contemporaneously.

On the other hand, it is well known that the efficiency is not the ultimate quantity, as what is more decisive for a turbine is the cost of the energy production; and not always minimum cost corresponds to maximum efficiency. Therefore, when considering a wind machine, there is another important criterion of optimization, that is to minimize the unit cost of the energy generation.

Moreover, which are the design parameters to consider for the WECS optimization analysis? It depends on the level of the analysis. In a general analysis, the WECS can be defined by the power curve, whose main data are the cut-in speed, the rated speed, the cut-off speed and the rated power. Simple models of the WECS power curve can be easily used, especially for pitch-controlled wind turbines (as the output power is variable in only one range of wind speed). Anyway, even though neglecting the cut-off speed, that has a minor effect, there are three design parameters to consider. The optimization can be carried out easily if the ratio of some of these parameters is assigned [1–3].

In a more specific analysis, many complex design parameters have to be considered: the hub height, the turbine diameter, the rotor speed, the aerodynamic shape of the blades, the rated power, etc. All these parameters affect both the energy output and the cost of the machine (directly and via the loads); evaluation and/or optimization of the energy unit cost [4] or of the total energy amount [5,6] is possible with suitable assumptions and algorithms.

In a general analysis, when matching the model of power curve of a WECS with the wind a model of the wind frequency distribution has to be assumed. Many analyses and investigations were carried out to find the best model, that is the model which gives less errors when substituted to the real distribution (histogram). The model more frequently examined are the lognormal, gamma function, Weibull and Rayleigh, and most authors [7–9] concluded that the Weibull distribution fits the real data better.

The Weibull distribution is a two-parameter model: the shape parameter k , that defines the shape of the distribution and mainly depends on the regularity of the wind regime,¹ and the scale parameter c , that is a measure of the mean wind speed, V_m . The Rayleigh distribution is a particular case of the Weibull distribution for $k = 2$, that is a value which the shape parameter is around in temperate climates.

For the sake of simplicity, some authors use the Rayleigh distribution [3,4,10,11]; however, this eliminates generality from the analysis and does not enable to take into account the local variations of the wind distribution. The shape parameter is widely varying not only from one site to another but also in the same site, where k varies with seasons, wind direction and years. The variation of k with the wind direction depends on different nature of the ground and related wind turbulence; the same takes place with the seasonal variation. The variation in the years depends not only on the climate casualness, which would be acceptable in a statistical analysis, but mostly on the climate modification, that appears when long-term data are analyzed [12,13]. The range of variation of k on a given site can be very wide and not negligible, especially the seasonal variation that can be larger than 50%. Moreover, when considering one value of k , an overall evaluation is carried out through a given period of time; however, as the energy is not a linear function of k , the overall energy can be very different from the real value that could be calculated with the variation of k with the time.

In any case, the Weibull distribution cannot be representative of every kind of wind regime; for instance, bi-modal maximum distributions [13] and non-zero wind calm distributions [14,15] are not fitted by the normal Weibull model.

Finally, a further question is: “what can be changed in the power characteristics of a WECS to adapt it to different local wind frequency distributions, maximizing the energy output in a different site?” As mentioned before, there are different levels of this kind of investigation. An analysis based on the matching of a model of power curve with the real wind distribution of a number of sites was presented formerly by the author [16–18]. The present analysis will be carried out on a more general level, that is by matching a simple model of the power curve of a WECS with the Weibull model of the wind frequency distribution. The aim of this work is to explore the effect of a change of site and/or season (that is of k and V_m) or of the hub height (that is of V_m) on the efficiency of a WECS and to evaluate the general design adjustments which would be needed to balance or reduce eventual efficiency losses.

¹The irregularity of the wind regime is measured by the standard deviation σ . The higher the regularity is, the lower σ and the higher k are. Mountainous areas and urban areas with irregular winds have values of k around 1 and 1.5, while trade-monsoon oceanic areas with very continuous winds have values around 2.5 and 4; in temperate coastal areas k ranges around 2. Moreover, in a given site k undergoes seasonal variations.

2. Analysis

The available power through an actuator disk is $P_a = 1/2 \rho A V^3$ and the available energy at constant temperature and density during the period T is

$$E_a = \int_T P_a dt = \frac{1}{2} \rho A \int_T V^3 dt = \frac{1}{2} \rho A T V_{m3}^3, \quad (1)$$

where $V_{m3} = ((1/T) \int_T V^3 dt)^{1/3}$ is the *root-mean-cube wind speed*. The output power of a wind generator can be written as

$$P = C_p \eta_m \eta_{el} P_a = \eta P_a = \frac{1}{2} \rho \eta A V^3, \quad (2)$$

where η_m and η_{el} are the efficiencies of the mechanical transmission and of the electric generation, while C_p is the power coefficient of the turbine. The function $P(V)$ is a characteristic of the WECS; it accounts for both aerodynamic efficiency and power control. The power efficiency of the WECS ($\eta = \eta_m \eta_{el} C_p$) is the ratio between output power and available power; among the three terms of η , C_p is the most variable (from 0 to $C_{p,max}$, that is around 0.4–0.5). The power law can be known only after field testing, however for general analysis, it can be simulated with suitable models. For pitch controlled blades, that are rather usual for medium-large size WECS, the models are based on three characteristic speeds (Fig. 1): V_c (cut-in), V_r (nominal or rated) and V_f (cut-off) and on the nominal or rated power P_r , while the rising curve can be well approximated with a parabolic law.² Therefore, the following model will be assumed, with different equation in the different ranges of wind speed:

$$\begin{aligned} \text{range } \{0\} \quad (0 \leq V \leq V_c) \text{ and } \text{range } \{3\} \quad (V_f \leq V \leq \infty): \quad & P = 0, \\ \text{range } \{1\} \quad (V_c \leq V \leq V_r): \quad & P = P_r \frac{V^2 - V_c^2}{V_r^2 - V_c^2}, \\ \text{range } \{2\} \quad (V_r \leq V \leq V_f): \quad & P = P_r = \frac{1}{2} \rho \eta_r A V_r^3. \end{aligned} \quad (3)$$

The rated efficiency is defined as

$$\eta_r = C_{p,r} \eta_{m,r} \eta_{el,r} = \frac{P_r}{P_{a,r}} = \frac{P_r}{\frac{1}{2} \rho A V_r^3} \quad (4)$$

that is known for a given machine. As $\eta = P/P_a$, the model gives the law $\eta(V)$ and it is easy to find [16,19] that the efficiency achieves a maximum within range {1} for $V = V_{opt} = \sqrt{3} V_c$, and it is

$$\eta_{max} = \frac{0.77 P_r}{\rho A V_c (V_r^2 - V_c^2)} = \eta_r \frac{V_r^3}{V_r^2 - V_c^2} \frac{1}{2.598 V_c} = \frac{\eta_r}{2.598} \frac{\phi^3}{\phi^2 - 1}, \quad (5)$$

where $\phi = V_r/V_c$ (design ratio). As the maximum power efficiency is achieved within the range {1}, the overall efficiency of the WECS is penalized if the machine does not

² It was shown by Power [1] that the quadratic relationship gives the best approximation as regards the energy evaluation.

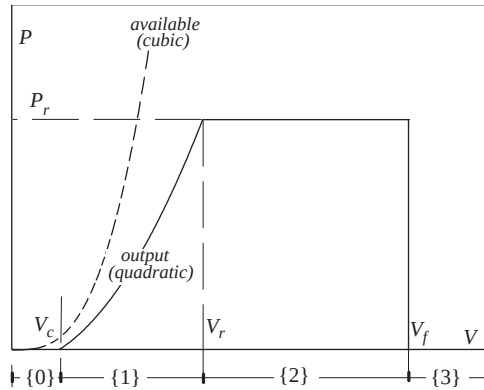


Fig. 1. Example of power curve of a pitch-controlled WECS.

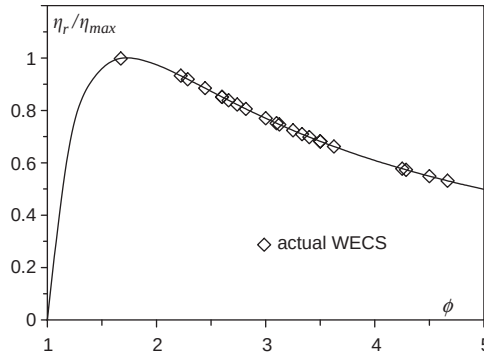


Fig. 2. Ratio between rated to maximum efficiency as function of the design parameter ϕ , with values of commercial WECS.

generate in this range. The present analysis considers as active both ranges {1} and {2}.

As in the design procedure $\eta_{\max}$ often can be considered rather constant (it is the best efficiency that can be achieved when designing a WECS and its variants), it is better to express η_r as function of $\eta_{\max}$.³

$$\frac{\eta_r}{\eta_{\max}} = 2.598 \frac{\phi^2 - 1}{\phi^3}. \quad (6)$$

Fig. 2 shows that the maximum of this function is achieved when $\phi = \sqrt{3}$, that is $V_r = V_{c,\text{opt}}$. For $\phi < 1.73$ the rated efficiency falls quickly down while for $\phi > 1.73$ the reduction is contained; that is high values of ϕ are conservative from the efficiency

³A discussion about η_r and $\eta_{\max}$ will be presented in the next section.

point of view. Actually, all examined WECS data (see next section) stay on the right of the maximum.

The rated power then can be expressed as

$$P_r = \frac{1}{2} \rho A \eta_r V_r^3 = 2.598 \frac{\rho \pi}{8} \eta_{\max} V_c^3 D^2 (\phi^2 - 1) = 1.02 \rho \eta_{\max} V_c^3 D^2 (\phi^2 - 1). \quad (7)$$

To calculate the output energy, the function $P(V)$ has to be integrated through the period of analysis T , and the contribution of the four ranges must be separated according to the probability density function $f(V)$. Let us define T_i the partial period of time (probable time) during which the wind speed falls within the range $\{i\}$ and f_i the respective fraction of the overall period T ; that is, omitting to consider ranges $\{0\}$ and $\{3\}$ which give no energy:

$$\begin{aligned} f_1 &= \frac{T_1}{T} = \int_{V_c}^{V_r} f(V) dV = \int_0^{V_r} f(V) dV - \int_0^{V_c} f(V) dV = C(V_r) - C(V_c), \\ f_2 &= \frac{T_2}{T} = \int_{V_r}^{V_f} f(V) dV = \int_0^{V_f} f(V) dV - \int_0^{V_r} f(V) dV = C(V_f) - C(V_r), \end{aligned} \quad (8)$$

where $C(V_j)$ is the value of the cumulative distribution function for $V = V_j$. The working contributions to the overall available energy within each range can be so calculated

$$\begin{aligned} E_1 &= \int_0^{T_1} P dt = \frac{P_r}{V_r^2 - V_c^2} \int_0^{T_1} (V^2 - V_c^2) dt = \frac{P_r}{V_r^2 - V_c^2} f_1 T (V_{m2,1}^2 - V_c^2), \\ E_2 &= \int_0^{T_2} P dt = P_r \int_0^{T_2} dt = P_r f_2 T, \end{aligned} \quad (9)$$

where $V_{m2,1}$ is the *root-mean-square wind speed* within range $\{1\}$:

$$V_{m2,1} = \left[\frac{1}{T_1} \int_0^{T_1} V^2 dt \right]^{1/2} \rightarrow \int_0^{T_1} V^2 dt = T_1 V_{m2,1}^2 = f_1 T V_{m2,1}^2. \quad (10)$$

Therefore, the overall output energy in the period T is

$$E = \int_0^T P dt = E_0 + E_1 + E_2 + E_3 = \frac{1}{2} \rho A T \eta_r V_r^3 \left[\frac{f_1 (V_{m2,1}^2 - V_c^2)}{V_r^2 - V_c^2} + f_2 \right]. \quad (11)$$

Now, let us define three parameters of merit of the WECS:

Plant utilization factor (or capacity factor), F_u : It is defined as the ratio between output energy and the nominal (or rated) overall energy $E_r = TP_r$:

$$F_u = \frac{E}{E_r} = f_1 \frac{(V_{m2,1}^2 - V_c^2)}{V_r^2 - V_c^2} + f_2. \quad (12)$$

Energy efficiency, η_T : It is the ratio between output energy and available energy:

$$\begin{aligned}\eta_T &= \frac{E}{E_a} = \eta_r \frac{V_r^3}{V_{m3}^3} \left[f_1 \frac{(V_{m2[1]}^2 - V_c^2)}{(V_r^2 - V_c^2)} + f_2 \right] \\ &= 2.598 V_c \eta_{\max} \frac{f_1(V_{m2[1]}^2 - V_c^2) + f_2(V_r^2 - V_c^2)}{V_{m3}^3}.\end{aligned}\quad (13)$$

Siting effectiveness, ε : It is defined as the ratio between output energy and $E_{\max} = \eta_{\max} E_a$, that is the energy exploitable if the WECS would operate constantly at the maximum efficiency $\eta_{\max}$:

$$\varepsilon = \frac{E}{E_{\max}} = 2.598 V_c \frac{f_1(V_{m2[1]}^2 - V_c^2) + f_2(V_r^2 - V_c^2)}{V_{m3}^3} = F_u \frac{\eta_r}{\eta_{\max}} \frac{V_r^3}{V_{m3}^3}.\quad (14)$$

The site effectiveness was adopted formerly by Pallabazzer [16–18]. Finally, the output energy and the energy efficiency can be calculated as

$$\begin{aligned}E &= F_u TP_r = \eta_T E_a = \varepsilon \eta_{\max} E_a, \\ \eta_T &= \varepsilon \eta_{\max}.\end{aligned}\quad (15)$$

These indices of efficiency characterize the interaction between wind machine and wind speed frequency distribution of the site. The decisive parameters of efficiency are the energy efficiency and the plant utilization factor; they can be and actually are used for comparison among WECS and among sitings.

The energy efficiency measures the level of exploitation of the local wind resource by a given wind machine, but it does not give any indication about feasibility and suitability of the siting: a given siting could have high energy efficiency but using little the power size of the machine, so the energy generation would cost too much. On the other hand, the plant utilization factor measures the energy that is produced with a given power size and, as the cost of the machine mainly depends on the power size, it can be also a rough measure of the economic suitability of the siting. This parameter is useful to compare WECS of same rated power (and approximately same cost), as in this case it is a measure of the output energy than can be achieved with that power size and cost. However, it does not give any indication about the efficiency of the exploitation: a given siting could have high economic suitability but exploiting little the wind resource, so it would be unsuitable from the ecological point of view because of poor resource saving.⁴ In practice, both parameters have to be considered in order to be certain that none is too small when privileging either according to the needs.

⁴ Ecological and not economical, because the energy efficiency is referred to the available wind energy, which is free (unlike the hydro energy and at least for now); therefore, this efficiency is significant only from an environmental point of view: a low value of energy efficiency means that the wind machine wastes a large part of the energy passing through its actuator disk, the environmental impact of which is well known.

On the other hand, while F_u is a purely kinematic parameter and can be used for comparison independently of the power efficiency of the WECS, η_T includes also the WECS efficiency. In order to obtain a purely kinematic expression of the energetic efficiency of the siting, the WECS efficiency was separated from the kinematic functions obtaining the site effectiveness. In the next section, a discussion about the use of these parameters will be presented with more detail.

In order to generalize the results, it is appropriate to introduce dimensionless parameters, using the mean wind speed V_m as reference; that is

$$v = \frac{V}{V_m}, \quad v_{m3} = \frac{V_{m3}}{V_m}, \quad v_c = \frac{V_c}{V_m}, \quad v_r = \frac{V_r}{V_m}, \quad v_f = \frac{V_f}{V_m}.$$

The foregoing quantities become

$$\begin{aligned} F_u &= f_1 \frac{(v_{m2,1}^2 - v_c^2)}{v_r^2 - v_c^2} + f_2, \\ \eta_T &= \eta_r \frac{v_r^3}{v_{m3}^3} \left[f_1 \frac{v_{m2,1}^2 - v_c^2}{v_r^2 - v_c^2} + f_2 \right] = 2.598 \eta_{\max} (\phi^2 - 1) \frac{v_c^3}{v_{m3}^3} F_u, \\ \varepsilon &= 2.598 v_c \frac{f_1 (v_{m2,1}^2 - v_c^2) + f_2 (v_r^2 - v_c^2)}{v_{m3}^3}. \end{aligned} \quad (16)$$

Finally, the probability functions will be computed by means of the Weibull model

$$\begin{aligned} f(v) &= k v^{k-1} \Gamma^k \left(1 + \frac{1}{k} \right) \exp \left[-v^k \Gamma^k \left(1 + \frac{1}{k} \right) \right], \\ C(v) &= 1 - \exp \left[-v^k \Gamma^k \left(1 + \frac{1}{k} \right) \right], \end{aligned} \quad (17)$$

where Γ is the gamma function. As it is known, the Weibull model, although never preferable to long-term recorded data, is a well approximate method for describing the local wind frequency distribution when the output energy of a siting has to be evaluated. This model is considered the best among many others [2,8] and is suitable when a wide comparative analysis among different sites and wind machines has to be carried out.

With this model, it is possible to calculate the siting parameters as functions of the shape parameter k only, which characterizes the local wind regime (see the Introduction). The fractions f_1 and f_2 are given by

$$\begin{aligned} f_1 &= C(v_r) - C(v_c) = \exp \left[-v_c^k \Gamma^{1/k} \left(1 + \frac{1}{k} \right) \right] - \exp \left[-v_r^k \Gamma^{1/k} \left(1 + \frac{1}{k} \right) \right], \\ f_2 &= C(v_f) - C(v_r) = \exp \left[-v_r^k \Gamma^{1/k} \left(1 + \frac{1}{k} \right) \right] - \exp \left[-v_f^k \Gamma^{1/k} \left(1 + \frac{1}{k} \right) \right] \end{aligned} \quad (18)$$

and the dimensionless square and cubic mean wind speeds are given by

$$v_{m2} = \frac{\Gamma^{1/2}(1 + 2/k)}{\Gamma(1 + 1/k)}, \quad v_{m3} = \frac{\Gamma^{1/3}(1 + 3/k)}{\Gamma(1 + 1/k)}. \quad (19)$$

However, to calculate the quadratic mean $v_{m2,1}$ within the range $\{1\}$ Eq. (19) cannot be used and we resorted to definition (10), that is

$$v_{m2,1} = \left[\frac{1}{T_1} \int_0^{T_1} v^2 dt \right]^{1/2} = \left[\frac{\int_{v_c}^{v_r} v^2 f(v) dv}{\int_{v_c}^{v_r} f(v) dv} \right]^{1/2}. \quad (20)$$

The integrals in Eq. (20) were calculated numerically for each value of k and each couple of $v_c - v_r$.

Apart from k , two main kinematic dimensionless parameters, both referred to the cut-in wind speed, were considered: the intensity ratio $\alpha = V_m/V_c \equiv 1/v_c$, and the WECS design ratio $\phi = V_r/V_c = v_r/v_c$.

3. Discussion on the parameters

As the aim of this work is to compare the efficiencies of a WECS in different sites, seasons or heights, with or without design adjustments, a criterion for the comparison has to be defined, although it is not very simple.

As Eq. (15) shows, the overall energy output E is measured by F_u at given rated power and by η_T at given available energy; therefore, F_u is mostly suitable for comparison among WECS with same rated power (e.g. same WECS in different sites) while η_T is mostly suitable for comparison among sites with same available energy (e.g. different WECS in the same site).

While η_T is clearly an energetic parameter, the nature of F_u is questionable. There is no problem when this parameter is considered as a ratio between two kinds of energy (output and nominal) both of which are variable depending on the site and on the WECS. But it is also used to consider F_u as a sort of feasibility parameter, since the cost of the WECS depends on the rated power and F_u can be seen as a sort of index of the unit cost of the energy. As it will be shown later on, this second significance of F_u is cause of misunderstandings and generally ought to be avoided.

As shown by Eq. (12), F_u is a purely kinematic parameter, that is it depends only on the characteristic speeds of the WECS and on the wind speed frequency distribution of the site, whereas η_T depends also on the machine efficiency. In order to evaluate the effect of the siting independently of the machine efficiency we introduced the site effectiveness: as η_T is proportional to ε through $\eta_{\max}$, ε can be used for comparison among sitings as long as $\eta_{\max}$ is the same. Generally, in the case of design adjustments we can consider $\eta_{\max}$ rather constant: this is correct if only the control is changed (same blades), but can be accepted also when the blades are to be changed (slightly different diameter and shape), as any design will be carried out with the same technology and procedure by the same manufacturer so as to achieve the maximum system efficiency. In case we have to compare different WECS (different manufacturers and/or different sizes), it is questionable whether $\eta_{\max}$ might be considered the same as technology and procedure are different, although it is around 0.4 for many but not for all WECS.

Table 1
Main data of some commercial WECS

No.	Model	H (m)	D (m)	P_r (kW)	V_c (m/s)	V_r (m/s)	V_f (m/s)	ϕ	η_r	$\eta_{\max}$
1	BONUS 55/11	24	16.3	55	4	13	28	3.25	0.200	0.276
2	BONUS 450	36	35	450	5	15	28	3	0.231	0.300
3	BONUS 600	40	44	600	3	15	25	5	0.195	0.391
4	BONUS 1 MW	45	54.2	1000	3	15	25	5	0.214	0.429
5	BONUS 1.3 MW	45	62	1300	3	15	25	5	0.213	0.426
6	BONUS 2 MW	60	76	2000	3	15	25	5	0.218	0.436
7	DOE-NASA MOD OA	40	38	200	5.8	9.7	18	1.67	0.322	0.323
8	ECOTECNIA 20/150	26	20	140	4	14.5	25	3.62	0.244	0.368
9	ENERCON E 33	35	33	330	3	14	25	4.67	0.234	0.441
10	GRUMMAN WINDSTREAM M25	12	7.6	20	4	12.5	22	3.12	0.376	0.504
11	INCLIN 1500		2.86	1.5	3.5	12	14	3.43	0.225	0.325
12	INCLIN 6000		4	6	3.5	12	14	3.43	0.460	0.664
13	HMZ WINDMASTER	30	25	300	5.5	15.5	25	2.82	0.273	0.339
14	JAMAHA WINDMEL	15	15	16.5	3.5	8	25	2.29	0.304	0.331
15	MITSUBISHI MWT 250	30	28	275	5	13	24	2.6	0.339	0.398
16	NORDIC 400	40	35.2	400	5	13	25	2.6	0.312	0.366
17	NORDTANK NTK 75F	23	16.6	75	4	18	25	4.5	0.099	0.180
18	NORDTANK NTK 130F	26	20.5	130	5	17	25	3.4	0.134	0.191
19	NORDTANK NTK 150F	26	20.5	150	4	17	25	4.25	0.154	0.267
20	NORDTANK NTK 300	31	28	290	5	15.5	25	3.1	0.211	0.281
21	PITCHWIND/PW20/14	24	14	20	3	9	30	3	0.297	0.386
22	PITCHWIND/PW25/14		14	25	3	9.5	30	3.17	0.316	0.427
23	PITCHWIND/PW30/14		14	30	3	10	30	3.33	0.325	0.458
24	PROVEN WT600		2.55	0.6	2.5	10	30	4	0.196	0.322
25	PROVEN WT2500		3.5	2.5	2.5	12	30	4.8	0.251	0.484
26	PROVEN WT6000		5.5	6	2.5	10	30	4	0.421	0.691
27	RIVA CALZONI M7	12	6.5	5.2	4.5	10	20	2.22	0.261	0.280
28	RIVA-CALZONI M30	33	33	200	4.5	11	20	2.44	0.293	0.331
29	VERGNET GEV26/220		26	220	5	14	25	2.8	0.252	0.311
30	VESTAS V20-100	24	20	96	4.5	15	25	3.33	0.151	0.213
31	VESTAS V25-200	29	25	220	3.5	15	25	4.29	0.221	0.386
32	VESTAS V27.225	30	27	225	4	14	25	3.5	0.239	0.350
33	VESTAS V39-500	39	39	500	4	14	25	3.5	0.254	0.373
34	WEST MEDIT 320	26	33	320	4.2	11.5	17.5	2.74	0.410	0.499
35	WEST GAMMA 60	66	60	1500	5	13.3	27	2.66	0.376	0.448

In order to compare different WECS and sitings, many authors [1–3,19–21], among which this author as well [16], assume constant $\eta_{\max}$ or η_r although there is no evidence of the validity of this assumption. In Table 1 a collection of data of 35 existing WECS is presented,⁵ and Fig. 3 shows the distribution of some parameters. As it can be seen, η_r and $\eta_{\max}$ vary from 0.1 to 0.46 and from 0.18 to 0.69, respectively, and there is no evidence of any concentration around a constant value,

⁵Data drawn from official publications and web sites. For some of these WECS a range of hub heights is available, but only the lower one was reported.

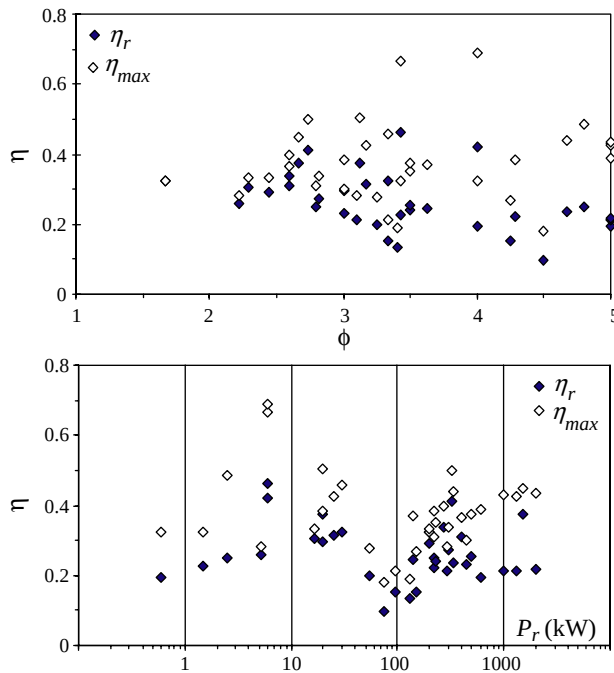


Fig. 3. Rated and maximum efficiency for some existing WECS.

independently of the power range. Therefore, it seems unreliable to compare different WECS assuming that the maximum or the rated efficiency are constant. This comparison should be accomplished only by multiplying the effectiveness by the maximum WECS efficiency. On the other hand, η_T allows the comparison of WECS of different power class, independently of the actual amount of available energy each of them collects. For instance, it can be useful for evaluating whether in a given site it should be better to install one big machine or many smaller ones for the same total power.

As a matter of fact, the behavior of both F_u and ε with the site and with the WECS characteristics will be analyzed and discussed; but one must bear in mind that the actual comparison among sitings requires known values of P_r and η_{max} .

Let us consider now two kinds of comparison: different WECS in the same site and same machine in different sites with or without design adjustment.

(a) *Different WECS in the same site*: When different WECS have to be compared in the same site both F_u (with energetic significance) and η_T can be used as they measure the output energy respectively at same rated power and same available energy, but the significance of the comparison is different. High F_u means that high output energy can be produced with a given power size; if known, the cost can be introduced for comparison. High η_T means high-energy output for given amount of available energy; as in a site E_a depends only on turbine diameter and hub height,

high η_T means low overall size to produce a given energy output, which is relevant for the environmental impact.

(b) *Same or modified WECS in different sites or with different hub heights:* If a given WECS without any modification is to be set up in different sites, F_u is a suitable parameter of merit both from the energetic and economic point of view, as both the rated power and the cost of the machine are the same (apart from differences of installation costs). In the case of comparison of the same wind turbine with different tower heights, the shape parameter k is the same but the mean wind speed V_m is different; however also the cost will be different, and F_u cannot be directly used with feasibility significance.

When hypothesizing a design adjustment (small change of the main parameters of a given WECS), we will consider only the possibility of modifying the nominal wind speed V_r and the nominal power P_r ; the cut-in wind speed depends on internal losses and cannot be easily changed and will be held constant; the effect of a variation of the cut-off wind speed is secondary and will be neglected; the maximum efficiency $\eta_{\max}$ will be considered independent of design adjustment. In this case, the rated power is proportional to $D^2(\phi^2 - 1)$, Eq. (7).

The rated wind speed V_r can be changed by modifying either the power control or the blades (shape and/or diameter). Changing only the control means that the range {2} (constant power) starts before or later and ϕ is changed; if the blades are not modified, the rated power will be different, according to Eq. (7). In this case, the comparison between two sitings by means of F_u is questionable. In fact, let us suppose a comparison between two sitings with lower and higher values of F_u . High value of F_u can be due to low nominal power (low nominal speed) rather than to high-energy output; in this case with two modified machines of same model we should get same or may be less output energy in the siting of higher F_u . From the energetic point of view it is still a measure of efficiency, but in this case we should make a mistake considering F_u a measure of economic suitability.

In the case of requiring P_r to be the same after a design adjustment for a different site, also the blades have to be modified, by changing the diameter in the ratio $(\phi^2 - 1)^{1/2}$ Eq. (7). Also in this case, F_u is not a suitable parameter of merit for economic comparison, as its definition does not account for changes of diameter Eq. (12). Higher F_u would mean that higher output energy can be obtained with a given rated power, which is a merit from the energetic point of view; however, in this case, although the rated power is the same, diameter and turbine cost are changed; therefore we achieve more energy but at different machine cost, and F_u cannot be used as index of unit cost. In Ref. [3], an economic evaluation was carried out for approaching this problem and allowing the comparison of different WECS on the basis of the unit energy cost; however, many assumptions were needed for the sake of simplicity; for instance, only the case $k = 2$ (Rayleigh model) and $v_c = 0.4$ ($\alpha = 2.5$) was examined.

In all these cases (same or modified WECS), as $\eta_{\max}$ can be considered rather constant, the site effectiveness is a better parameter of merit and can be preferable to F_u .

About the procedure of analysis the following remarks have to be made:

- (i) for a given WECS, a change of mean wind speed means change of α while ϕ keeps constant; however, small variations of ϕ are considered (design adjustment);
- (ii) in a given site, a change of hub height means a variation of mean wind speed (that is of α) at constant shape parameter k ;
- (iii) a change of k and/or of α at constant ϕ is what takes place in the siting of a given WECS during seasonal or yearly variations and/or wind direction variations;
- (iv) F_u and ε vary from 0 to 1; values of F_u higher than 0.5 are rare; usually the siting is considered unsuitable for $F_u < 0.2$;
- (v) the main parameters will be α (measure of the mean wind speed) and ϕ (WECS internal design parameter); the suitable range of ϕ is 1.5–5; the operative range for α will be 1–4 (we will not consider the case $V_m < V_c$); the cut-off wind speed affects very little the results as it is high enough to allow neglecting the lost energy: a fixed value $v_f = 5$ was assumed all throughout.

4. Results

4.1. Plant utilization factor

Fig. 4 shows the effect of the shape parameter k on F_u : the curves $F_u(k, \alpha)$ are plotted for some values of ϕ . Each diagram describes what happens to a given WECS without any modification ($\phi = \text{constant}$) owing to a change of climatic area (k) and of mean wind speed (α); the latter effect could be due to variations of hub height.⁶ The effect of α is always beneficial, as expected, but that of k depends on the value of both ϕ and α , being penalizing in some cases and improving in others: for low design ratios ($\phi = 2\text{--}3$), F_u increases with k when the mean wind speed is high ($\alpha = 2\text{--}3$) but decreases when V_m is low ($\alpha = 1\text{--}1.5$); for high design ratios, the penalizing values of α extends to higher values of α (2–2.5). Generally, it can be said that the higher F_u is the more intense the improving effect of k is; while initial low values of F_u get worse by any increase of wind regularity. The effect of k is negligible only for intermediate values of α , that is: $\phi = 2, \alpha \cong 1.5$; $\phi = 3, \alpha \cong 2$; $\phi = 4, \alpha \cong 2\text{--}2.5$; $\phi = 5, \alpha \cong 2\text{--}3$. This means that the assumption $k = 2$ (Rayleigh model) that is made in many analyses (see the Introduction) is acceptable only within these ranges of ϕ and α , as out of them the effect of a variation of k on F_u can be higher than 30–40%.

⁶The function $F_u(\alpha)$ cannot be used directly for evaluating the effect of the hub height. When supposing to increase the hub height, V_m and α increase, but the function $F_u(\alpha)$ does change too, due to the shifting of the frequency distribution to higher speed: the available energy increases with the height but the output energy increases little less as a larger part of the available energy (beyond the rated speed) is cut by the power control.

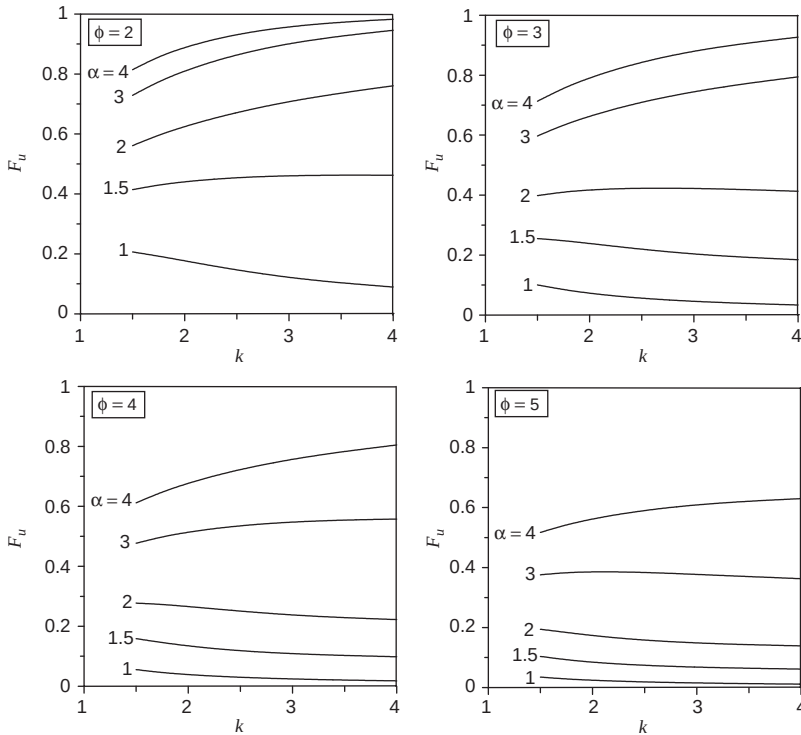


Fig. 4. Plant utilization factor as function of k and α , at constant design ratio ϕ .

Therefore, the suitability of a WECS to different climatic sites depends on the values of both ϕ and α : if no design adjustment is considered ($\phi = \text{constant}$) it depends on the local mean wind speed (α).

The effect of ϕ is always penalizing, as it can be seen from Fig. 5a, where $F_u(\phi, k)$ is plotted for two values of α . For all values of k and α , when ϕ is increased the plant utilization factor diminishes. Indeed, increase of ϕ implies increase of rated speed V_r and rated power (Eq. (7)) and decrease of F_u . However, as it concerns the output energy, from Eqs. (7) and (15) we have:

$$E = F_u TP_r = 1.02 T \eta_{\max} V_c^3 D^2 F_u (\phi^2 - 1) = \text{const. } F_u (\phi^2 - 1),$$

where the constant means that we are considering a WECS of given diameter and given maximum efficiency. It can be seen in Fig. 5b that the behavior is completely different and E actually is increased by any increase in ϕ , because the increase in $(\phi^2 - 1)$ is always much higher than the respective reduction in F_u . Therefore, when this kind of design adjustment ($D = \text{const.}$ and $\eta_{\max} = \text{const.}$) have to be considered, the diagram of Fig. 5a cannot be directly used for comparison and that of Fig. 5b is needed.

The diagrams of Fig. 6 show that the benefic effect of α is different depending on the design ratio, being more efficacious at low than at high values of ϕ ; for $\phi = 2$, the

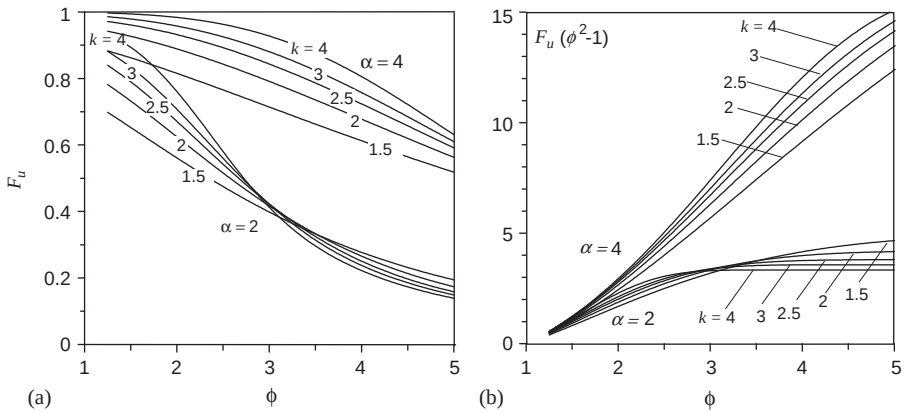


Fig. 5. (a, b) Effect of the design ratio ϕ on the plant utilization factor and on the output energy.

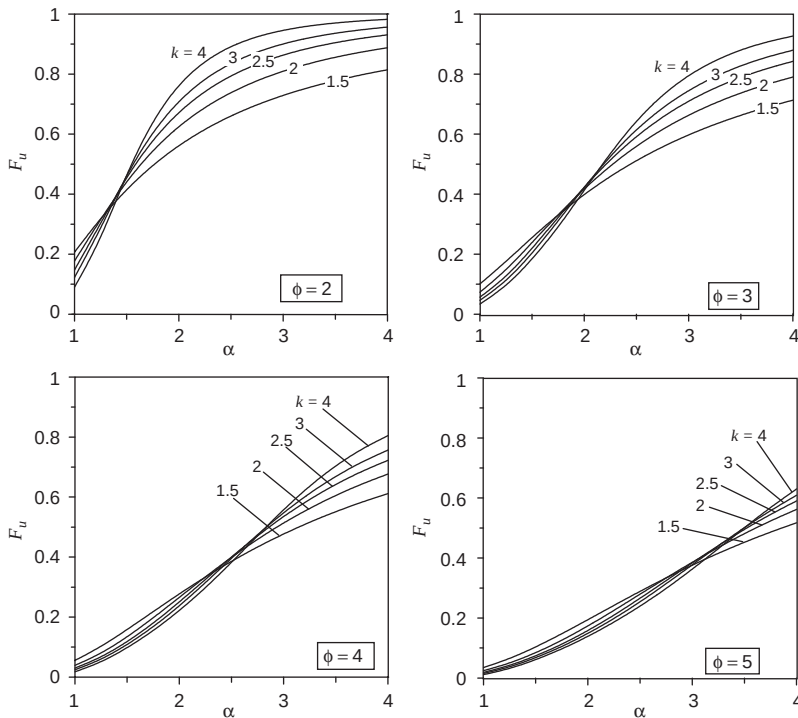


Fig. 6. Plant utilization factor as function of α and k , at constant design ratio ϕ .

plant utilization factor is practically always higher than 0.2, while for $\phi = 4$ and 5 it is lower than 0.2 up to $\alpha = 2$. The effect of the design parameter ϕ is mainly to squash down the curves $F_u(\alpha)$; that is at same V_m (e.g. same site) a WECS with higher ϕ achieves lower F_u , while for WECS with higher ϕ the same value of F_u is

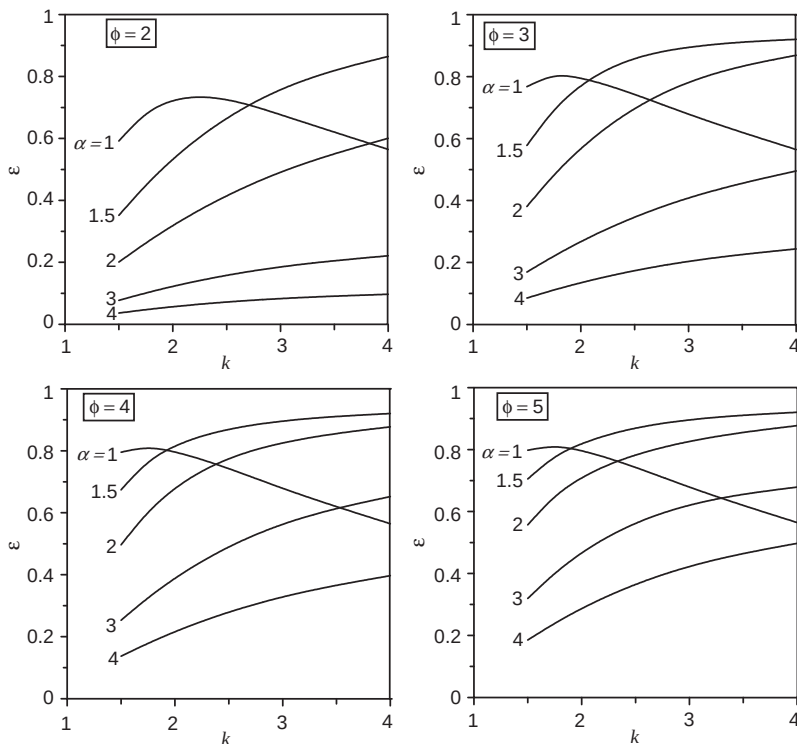


Fig. 7. Site effectiveness as function of k and α , at constant design ratio ϕ .

achieved under higher mean wind speed; for $\phi = 5$, F_u can be hardly higher than 0.5, while for $\phi = 2$, $F_u > 0.5$ in a wide range of α . It can also be seen that for given ϕ the curves $F_u(\alpha)$ cross and the k -effect is reversed, as already shown commenting Fig. 4.

However, at low ϕ , the improvement due to increase in the mean wind speed is intense for small α , becoming less efficacious when α is high, while at high ϕ the effect is quite uniform in all the range of α . Also the effect of k becomes feeble as ϕ increases.

4.2. Site effectiveness

As shown in Fig. 7, the site effectiveness is more k -dependent than F_u is. Except for the lowest α ($= 1$), that shows a maximum, ε is monotonically increasing with k . Therefore, $\eta_{\max}$ being constant for a given machine, the energy efficiency of a WECS is always improved by k (i.e. the wind regularity): a given mean wind speed allows more wind energy exploitation when the wind regularity is high. This effect is well known, but the analysis shows that it can be very intense, as for instance ε can double when k passes from 1.5 to 3.

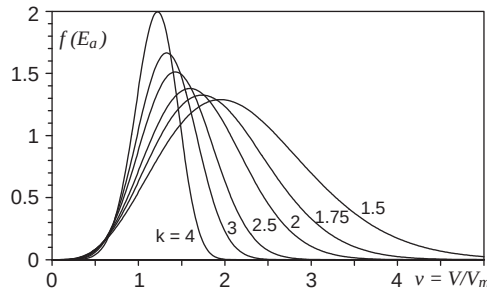


Fig. 8. Wind energy probability distributions according to the Weibull model.

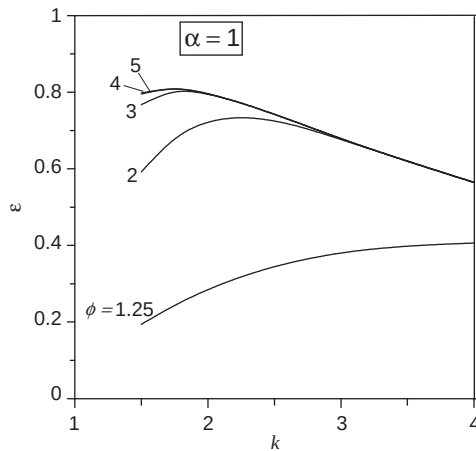


Fig. 9. Site effectiveness as function of k for very low mean wind speed.

The eventuality of a maximum depends on the shape of the energy distribution curve (see Fig. 8): when k increases the curve contracts on the left⁷ and two opposite effects take place: (i) a part of the wind energy falls within the rising part of the WECS power curve (range $\{1\}$) and can be exploited at high power efficiency, and (ii) the fore part of the wind energy curve shifts before the cut-in speed and is not exploited. At enough high values of α , the former effect prevails. The case of very low mean wind speed can be seen in Fig. 9 for $\alpha = 1$: apart from the lowest value of ϕ, ε shows a maximum then it decreases when k increases. This is because in this case most part of the energy falls at wind speeds lower than the cut-in speed, as can be seen in Fig. 10 for extreme values of k : when k increases the maximum approaches the cut-in wind speed (that is $= V_m$ for $\alpha = 1$) and leaves an increasing part of the

⁷For k passing from 1.5 to 4 the wind speed of maximum energy passes from $1.95V_m$ to $1.24V_m$.

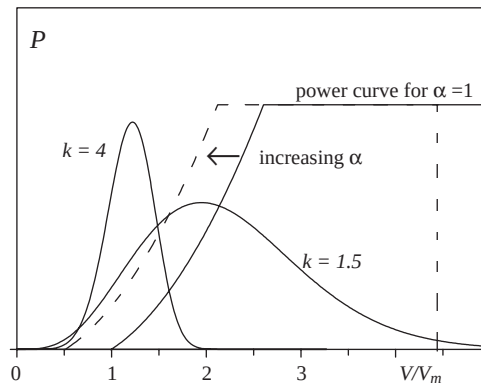


Fig. 10. Effect of k (extreme values) on a power curve with $\alpha = 1$; when α increases the power curve moves leftward.

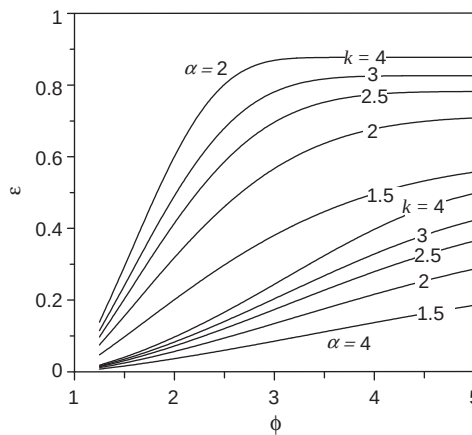


Fig. 11. Effect of the design ratio ϕ on the site effectiveness.

energy distribution curve before the WECS power curve. When α increases, the power curve shifts to left and “gulps down” the wind energy distribution.

Fig. 11 shows the effect of the design ratio ϕ for two values of α ; this effect is monotonically improving as well, because when the rated speed increases a larger part of the energy curve falls within range $\{1\}$, where the power efficiency is higher. At lower α the improvement is mostly concentrated within $\phi = 2$ – 3 , while it becomes more uniform as α increases.

In Fig. 12 the function $\varepsilon(\alpha, k)$ is plotted for some values of ϕ , and a maximum is shown in all cases. This behavior is due to two opposite effects of increasing α : the reduction in V_c allows the capture of low-speed energy fractions so increasing the output energy, while the increase in V_m raises the available energy. Increasing k results in increasing both $\varepsilon_{\max}$, drawing it toward higher α , and α_{opt} , that is: regular

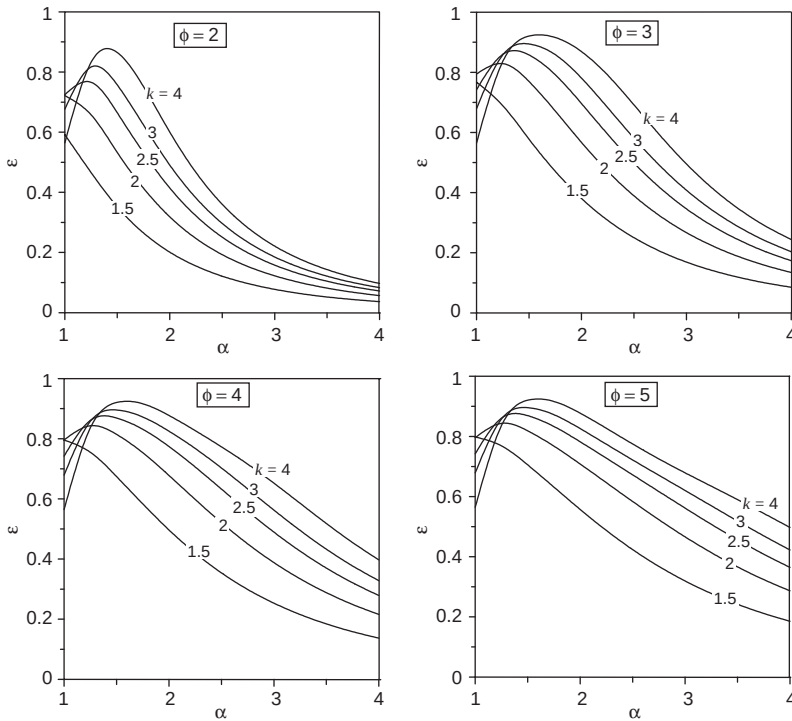


Fig. 12. Site effectiveness as function of α and k , for constant design ratio ϕ .

wind regimes improve the energy exploitation but need high values of mean wind speed to achieve maximum efficiency.

The existence of a maximum restrains the suitable values of α and ϕ , as ε falls down when these parameters deviate from the optimum. However, although the maximum could be very high, exceeding 0.9, it takes place at too low values of α , not higher than about 1.5; this means that excluding the range $\alpha < 1.5$ the site effectiveness actually is a decreasing function of the mean wind speed. This behavior is very different from that shown by F_u , and happens because in this range the available energy increases more than the output energy. In the range $\alpha > 1.5$ the effect of k is monotonically improving: at constant α and ϕ the site effectiveness can increase by up to 0.5 when k passes from 1.5 to 4. In any case, it is unlikely to achieve the maximum site effectiveness in irregular wind regimes, because α_{opt} is very low at low k ; it can be easier to achieve the maximum when $k > 2$. When ϕ increases there is a small increase of α_{opt} while ε_{max} is practically the same.

4.3. Comparison between ε and F_u

As the effects of k and of α on F_u and ε are different, it will be useful to plot both F_u and ε on the same diagram, in order to summarize the comparison. Fig. 13 shows this graphical coupling for two values of the design ratio ϕ (2 and 4) and of α (2 and 4).

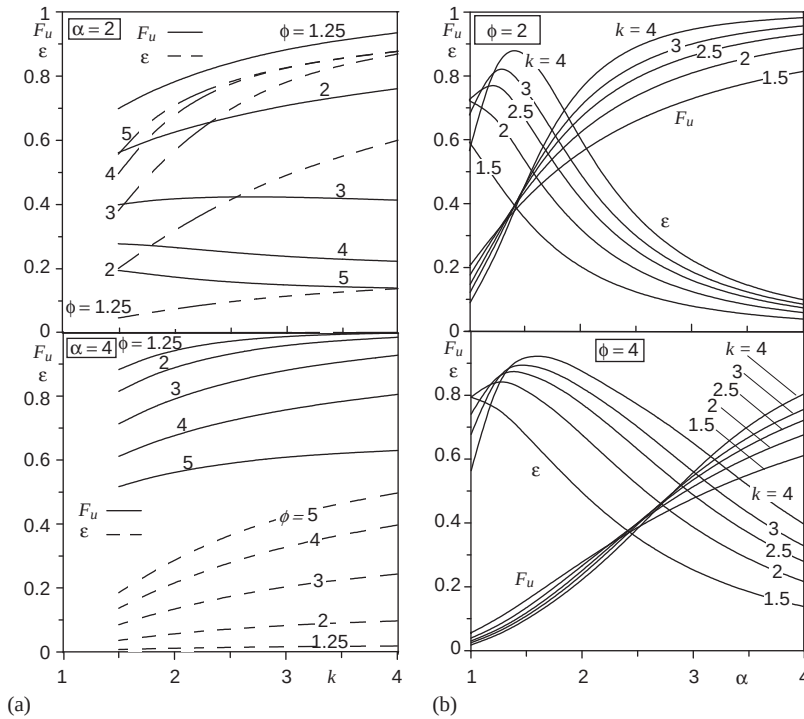


Fig. 13. (a, b) Comparison between plant utilization factor and site effectiveness for given WECS.

From Fig. 13a, it can be seen that: (i) increase of ϕ penalizes F_u and improves ε , and (ii) increase of k improves ε while it improves or penalizes F_u depending on the values of ϕ and α . The opposite effects of ϕ on F_u and on ε are due to the fact that when ϕ increases the energy output increases as well, but F_u is referred to the rated power, that increases with ϕ^3 , while ε is referred to the available energy that does not depend on ϕ . About the k -effect, ε shows more dependence on the wind frequency distribution than F_u .

Fig. 13b shows that F_u as function of α becomes small just in the range where the efficiency achieves its maximum, and this effect is intensified when ϕ is high and k is small. Practically, for $k < 2$ it is almost unsuitable to work in the range of low α (low mean or high cut-in speed) as F_u becomes too low. In most practical cases ($\alpha > 1$ –1.5), increasing α results in reducing ε . This behavior is very different from that of F_u , which is strongly improved by any increase in α . This is because, considering a given WECS (that is constant P_r and ϕ): (i) F_u (Eq. (12)) depends only on the energy output E , which increases when V_c decreases, while (ii) ε (Eq. (14)) and η_T (Eq. (13)) depend also on the available energy E_a , which also decreases when V_m decreases. As matter of fact, when α increases, F_u says that the energy supplied by a given WECS increases, while ε says that the exploitation of the wind energy takes place with lower energy efficiency.

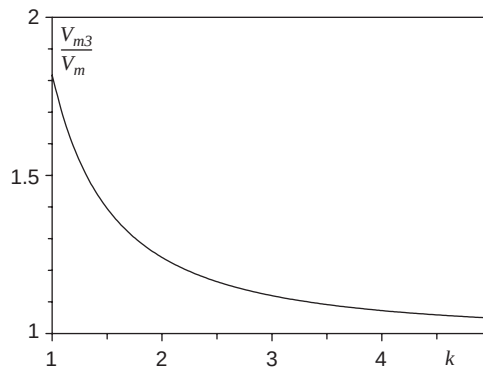


Fig. 14. Effect of the shape coefficient on the dimensionless root-mean-cube wind speed.

Let us consider two simple cases.

(1) Effect of a variation of the hub height on given wind turbine. In this case k , V_c , V_r and ϕ are constant, while α increases when increasing the tower height. The effect of α can be seen on the k -constant curves at constant ϕ (Fig. 13b): when α increases F_u increases and ε decreases (in the most common case, i.e. $\alpha > 1$ –1.5). In fact: if the mean wind speed increases, the energy output increases, and, as the rated power is the same, F_u increases too; but, as the available energy also increases with the cubic law, ε will increase or decrease depending on the value of α and ϕ , but more frequently decreases, as the cubic law is dominant. So, rising the tower we get more energy output with lower energy efficiency.

(2) Effect of a variation of site with same mean wind speed on given WECS. In this case V_c , V_r , V_m , α and ϕ are constant, while k varies. It can be seen from the ϕ -constant curves at constant α (Fig. 13a) that, owing to the increase in k , ε in most cases increases while F_u will increase or decrease depending on the values of α and ϕ . This is because ε depends on the available energy, which at same mean wind speed decreases when the wind regularity increases (see Fig. 14), while F_u depends only on the energy output and is affected by the shape of the wind frequency distribution on the power curve of the WECS. So, setting up the WECS in a site with higher k we can collect higher or lower output energy depending on the WECS, but in any case the efficiency of the energy exploitation will be better.

It is clear that dimensionless parameters of merit like F_u and ε can be used as a direct measure of the output energy only if the reference energy is the same; that is, to compare sitings respectively with same rated power (i.e. same WECS) and with same available energy. The first case is simple as it means to compare the output of a given WECS in different sites and/or with different hub heights. The second case is less immediate, as the available energy depends on the site, the hub height and the turbine diameter. Actually, through η_{ma} , ε is the measure of the efficiency of the conversion and allows an energetic comparison among wind machines of any power class and diameter, but for practical use it is less immediate than the plant utilization factor. In all other cases, the comparison requires the calculation of the output energy by means

of Eq. (15), and either F_u or ε can be used for this aim. The plant utilization factor is more practical when design adjustments of a given WECS for different sitings and/or hub heights have to be evaluated; the site effectiveness is more significant when one has to compare the output of different WECS in the same site.

5. Conclusions

The effect of the wind climatic characteristics on the WECS energy output has been analyzed by means of two efficiency parameters, the plant utilization factor and the energy efficiency. It has been shown that neither the rated nor the maximum efficiency of the WECS can be assumed as constant in a general analysis, as the examination of the data of 35 commercial WECS shows that there is no basis for this hypothesis. The maximum efficiency $\eta_{\max}$ can be reasonably assumed as constant only when the design adjustment of a given WECS is considered. In all other cases, the comparison among different WECS needs to pass through the WECS data.

It has been shown that the use of the efficiency parameters F_u and ε with a view of comparison must be made carefully: it is immediately significant if the efficiency of a given WECS in different sites and/or hub heights has to be examined but is deceptive when comparing each other different WECS or same WECS after a design adjustment.

The analysis of the effects of the Weibull shape parameter k and of the dimensionless mean wind speed α on the efficiency parameters F_u and ε gives the following main results:

(1) The shape parameter k has always a beneficial effect on the site effectiveness, that strongly increases with it, while affects differently the plant utilization factor: for low values of the design ratio ϕ (≤ 3) increasing k results in decrease or increase of F_u depending on α being low (≤ 1.5) or high (> 2); for high values of ϕ the decreasing effect extends. Considering a given WECS, the output energy can be either improved or penalized by the increase of k depending on α and ϕ , while the energetic efficiency of the conversion is always improved.

(2) The effect of k on F_u can be neglected only for intermediate values of α ($\phi = 2, \alpha \cong 1.5$; $\phi = 3, \alpha \cong 2$; $\phi = 4, \alpha \cong 2-2.5$; $\phi = 5, \alpha \cong 2-3$); beyond these ranges the variation of F_u through the full range of k can be up to 30%; this means that the indiscriminate use of the Rayleigh model is questionable.

(3) The relative mean wind speed α has always a beneficial effect on the plant utilization factor, while the site effectiveness shows a maximum at very low values of α (< 1.5): in the practical range of α , increasing α results in a reduction of the site effectiveness and of the energy efficiency. Considering a given WECS, the increase of α is what happens if the hub height is increased or the WECS is set up in the same climatic area but more energetic site: in these cases, the output energy is improved while the efficiency of the conversion is reduced.

(4) Also the effects of the design ratio ϕ are opposite: increasing ϕ results in reduction of F_u and improvement of ε ; as variation of ϕ implies variation of the rated power, in this case we cannot consider a given WECS but only a modified WECS;

it was shown that increasing ϕ improves both the energy efficiency and the output energy.

As a final comment, answering to the question put in the Introduction, we can say that there is no possible maximization for the plant utilization factor, which is monotonic function of the design parameters ϕ and α ; practically, the same occurs for the energy efficiency, as its maximum take place at too low values of α . Both efficiency parameters can be improved with design modifications but never maximized.

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